

UNIVERSITY OF THE FREE STATE
DEPARTMENT OF COMPUTER SCIENCE AND INFORMATICS
CSIQ1614/CSIS1614

DATE: 14 May 2026

MARKS: 87

ASSESSORS: Mr. T. LESESA and Mr. J. VIEIRA

MODERATOR: Prof. T. Stott

TIME: 2H 30M (15M for Submission) minutes

-
- All files must include your name and student number at the very top of every file. We subtract marks for every missing occurrence **(2 marks)**.
 - Your program must run **(2 marks)**.
 - We are blind to green. We don't mark the code that is commented out.
 - This is a **closed-book** test.
-

This practical comprises of three sections:

- Theory quiz (Section A) – 15 marks
- Code comprehension (Section B) – 15 Marks
- Visual Studio project (Section C) – 57 Marks

The instructions for Section C are outlined below. Refer to the quizzes on Blackboard for instructions on Section A and Section B.

SECTION C (57)

- Add a comment block with your student number, name and surname, today's date and the name of the question at the top of every file.
- Adhere to all programming and naming conventions as taught and ensure that your code is aligned properly. Do not add excessive white space.
- You do not have to add any defensive programming unless you are requested to.
- Ensure that at the end of every question the instruction to the user that he/she should press any key to exit the application, followed by a statement that ensures that the console window is displayed until the user presses a key.
- This is not an open-book test. You may only use the internet to access Blackboard for opening this document and submitting your work.

Question 1

Monthly Expense Tracker

Using Visual Studio, create a C#, .NET Framework console application, and name your project SemesterTest2.

Instructions:

You are required to write a C# program that records and analyses monthly expenses. The program must allow the user to enter multiple expense categories, calculate the total expenses, and display a complete monthly expense report. All monetary outputs should display to two decimal places.

The program must adhere to the following requirements:

Main **[17]**

The **Main()** method is the entry point of this program, and it is responsible for the overall control and flow of the application.

- Print the program's welcome heading message.
- Prompt the user to enter the number of expense categories they would like to process and save the number.
- Declare two arrays: one for category names and one for expense amounts, both use the number of expense categories entered by the user.
- Create a counter-controlled loop, that will process the number of expense categories entered by the user.
- Inside the loop:
 - Print the category heading, include the category count.
 - Call the **GetExpenseDetails()** method and pass the current index and the two arrays by reference along with the call.
- Outside the loop, call the **DisplayReport()** method and pass the two arrays and the number of categories.
- Prompt the user to "Press any key to exit..." and wait for the keyboard input.

Get Expense Details**[16]**

The **GetExpenseDetails()** method is used to get the expense category name and amount spent from the user. The method accepts the current index and the two arrays by reference.

- Prompt the user to enter the expense category name.
- Prompt the user to enter the amount spent.
 - Use try-catch for defensive programming, to handle invalid amount input.
 - If the amount is negative or invalid, set the amount to zero (0) and print appropriate messages to tell the user that the amount is negative or that the amount is invalid.
 - If the amount was correct, store the values in the arrays at the given index so that it can be updated in the main method.

Display report**[24]**

The **DisplayReport()** method is used to display the complete monthly expense report to the user. The method accepts two arrays and the number of categories as parameters.

- Print the report heading.
- Use a counter-controlled loop to go through all categories.
- Display each category name and its amount.
- Calculate and display the total monthly expenses.
- Use a conditional statement to display suitable performance feedback based on the total expenses:
 - When the total expenses are more than 15000, print the message "**Warning: High spending this month! Please review your expenses.**"
 - When the total expenses are more than 8000, print the message "**Moderate spending detected.**"
 - For all other cases, print the message "**Good expense control! Keep it up.**"

2 marks are reserved for naming conventions and code formatting as taught.

[57]

NB. Remember to follow the submission instructions.

Sample Outputs:

```
Welcome to the Monthly Expense Tracker!

How many expense categories do you wish to process? 3

--- Category 1 ---
Enter expense category: Internet
Enter amount spent: RSeven Hundred

Invalid amount entered. Recorded as 0.

--- Category 2 ---
Enter expense category: |
```

Figure 1: Invalid amount input

```
Welcome to the Monthly Expense Tracker!

How many expense categories do you wish to process? 3

--- Category 1 ---
Enter expense category: Internet
Enter amount spent: R699

--- Category 2 ---
Enter expense category: DStv
Enter amount spent: R450

--- Category 3 ---
Enter expense category: Airtime
Enter amount spent: R200

===== MONTHLY EXPENSE REPORT =====
Internet           : R699,00
DStv               : R450,00
Airtime            : R200,00
-----
Total Monthly Expenses : R1 349,00
Good expense control! Keep it up.

Press any key to exit...
```

Figure 2: Good monthly expense control

```
Welcome to the Monthly Expense Tracker!

How many expense categories do you wish to process? 4

--- Category 1 ---
Enter expense category: Rent
Enter amount spent: R8500

--- Category 2 ---
Enter expense category: Groceries
Enter amount spent: R3200

--- Category 3 ---
Enter expense category: Transport
Enter amount spent: R1800

--- Category 4 ---
Enter expense category: Electricity
Enter amount spent: R1450

===== MONTHLY EXPENSE REPORT =====
Rent                : R8 500,00
Groceries            : R3 200,00
Transport            : R1 800,00
Electricity          : R1 450,00
-----
Total Monthly Expenses : R14 950,00
Moderate spending detected.

Press any key to exit...
```

Figure 3: Moderate monthly expense detected

END OF QUESTION PAPER

TOTAL = 87

Seat Number:

CSIQ/CSIS1614 (Semester Test 2 - 2026): Submission Verification Slip

I, _____, _____ hereby verify that:

(Initials and surname)

(Student number)

1. I have included **all** my Section C files in my ZIP folder.
2. I have uploaded the **correct** ZIP folder to Blackboard.
3. I have successfully **completed the submission** procedure on Blackboard.
4. I have **downloaded** my zipped folder and verified that I have submitted the correct files.

Signature – Student

Signature - Invigilator

